Unlike the manually controlled or GPS-assisted systems of the second generation, third-generation drones feature fully digital flight control architectures with redundant navigation systems, real-time data links, and autonomous decision-making algorithms that enable adaptive mission execution. They are designed for diverse operational roles, including intelligence, surveillance, and reconnaissance (ISR), target acquisition, precision strike missions, and scientific or civilian applications such as mapping and disaster response.

These UAVs typically belong to the medium-altitude long-endurance (MALE) and high-altitude long-endurance (HALE) categories, offering extended flight durations of over 24 hours and operational ranges exceeding 1000 kilometers. Equipped with advanced payloads such as electro-optical/infrared (EO/IR) sensors, synthetic aperture radar (SAR), and signal intelligence (SIGINT) systems, they provide continuous, high-resolution situational awareness in real time.

4.1.2 Basic characteristics of third generation of drones

The third generation of drones represents a technological milestone in the evolution of unmanned aerial vehicles (UAVs), embodying the shift from remotely controlled reconnaissance systems to intelligent, networked, and multi-role aerial platforms. These drones combine autonomy, endurance, precision, and connectivity, enabling them to perform complex missions ranging from long-range surveillance to precision strikes and civilian data-gathering operations. Their design and functionality reflect major advances in aerodynamics, communication networks, onboard computing, and sensor integration.

Below are the key characteristics that define third-generation drones: